

Data Science
PORTFOLIO

Mochammad Fahd Ali Hillaby | 2026



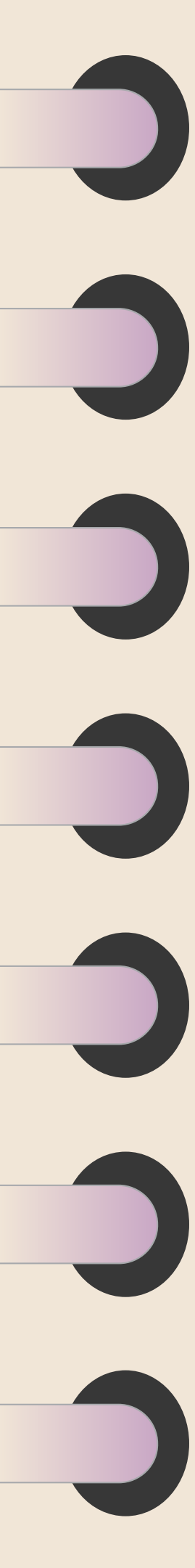
M. FAHD ALI HILLABY

Data Science Technology Student

Hello, my name is Aby

I am currently pursuing a fresh graduate in Bachelor's degree in Data Science Technology. I have a strong interest in the fields of data analysis and statistics. I am passionate about leveraging the potential of data and advanced technology to address complex challenges, uncover valuable insights, and develop innovative solutions. Through my academic journey, I aim to deepen my understanding of data-driven approaches and contribute to impactful projects that make a difference in various industries.

Mochammad Fahd Ali Hillay | 2026



EDUCATION

Bachelor of Data Science Technology

Airlangga University
2022 - now



ACTIVITIES

BEM FTMM 2023

Dirjen Arttaiton Medkominfo
2023

BEM FTMM 2024

Inspectorat General
2024

IRIS 2024

Researcher
2024

IRIS 2025

Head of Public Relation
2025



EXPERIENCE

PT Semen Indonesia

CSR Data Analyst Intern
2026

PT Indospring, tbk

Human Capital Data Analyst
2025

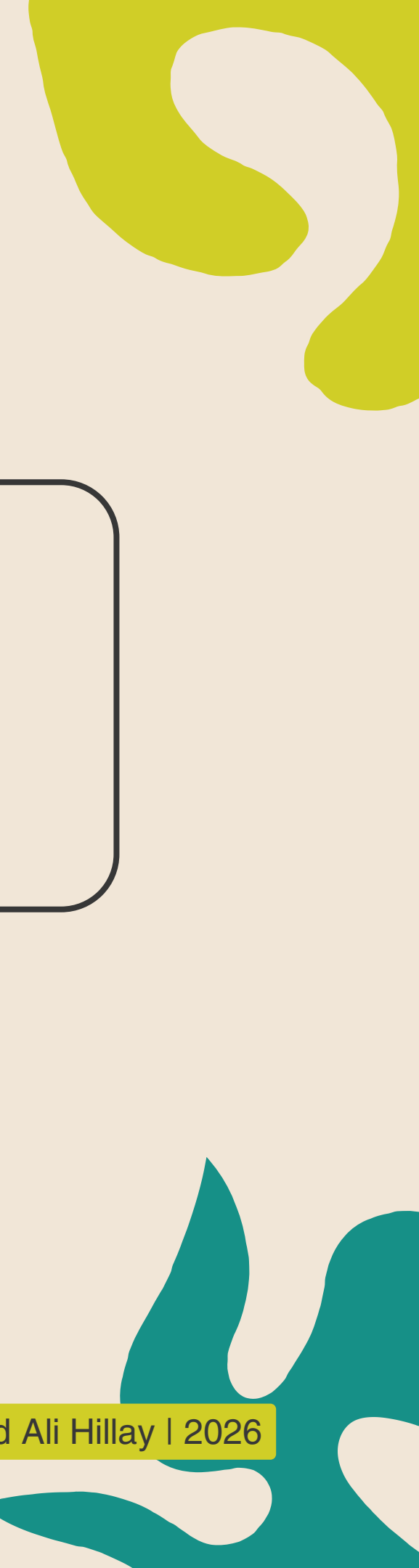
ICATAM 2024

Contribute Paper
2024



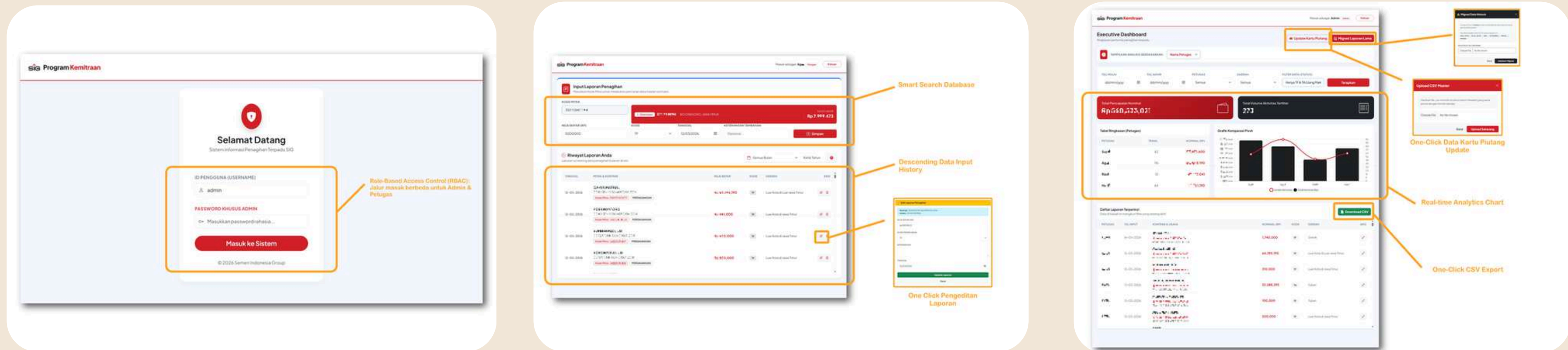
DATA ANALYSIS PROJECT'S

2023-2026



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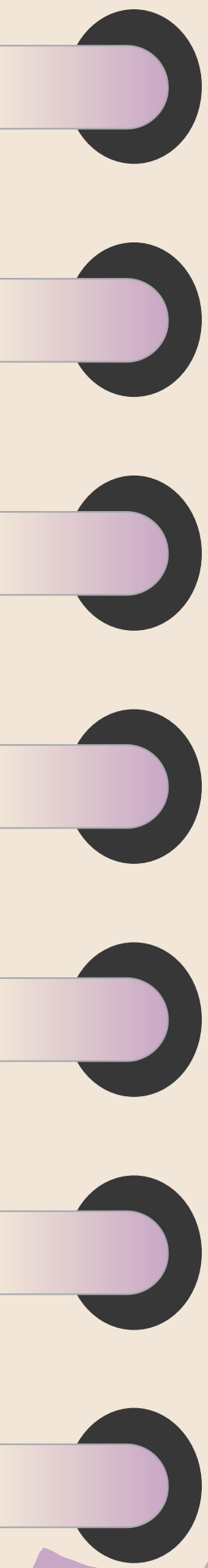
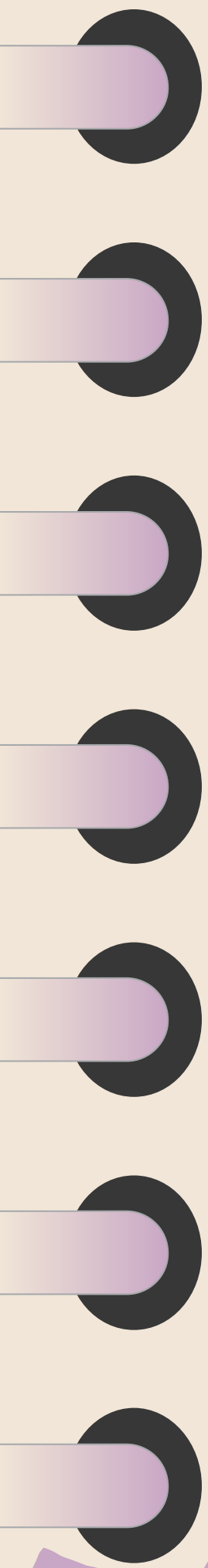
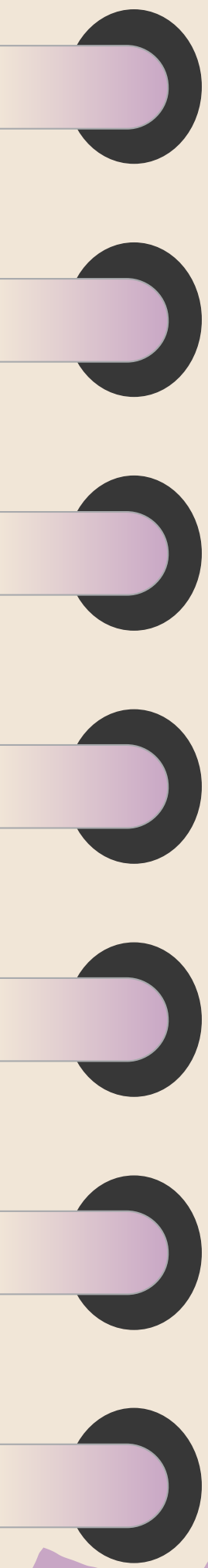
Project Overview: Integrated CSR & COMDEV Reporting System



A full-stack web application designed to digitize and centralize the reporting process for CSR and Community Development programs at PT Semen Indonesia (Persero) Tbk (SIG). Built using Next.js and a serverless PostgreSQL database, this system replaces traditional, fragmented spreadsheet-based workflows.

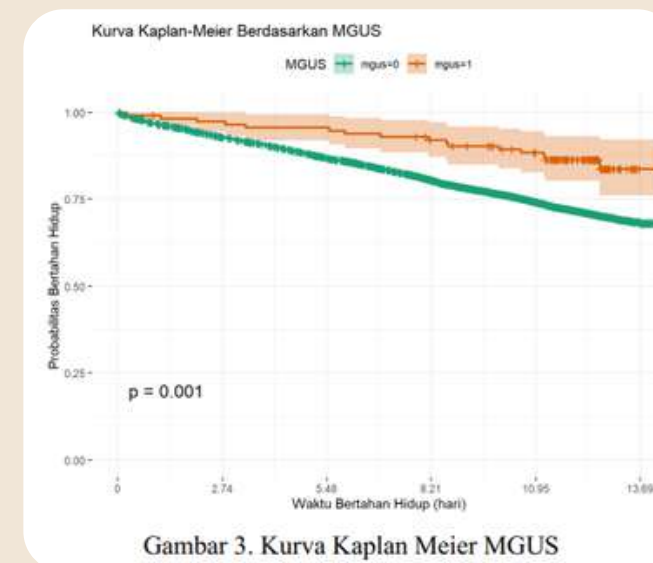
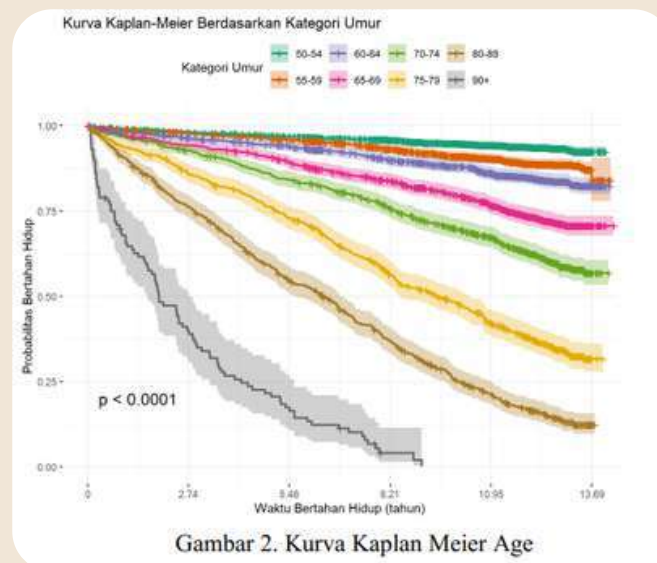
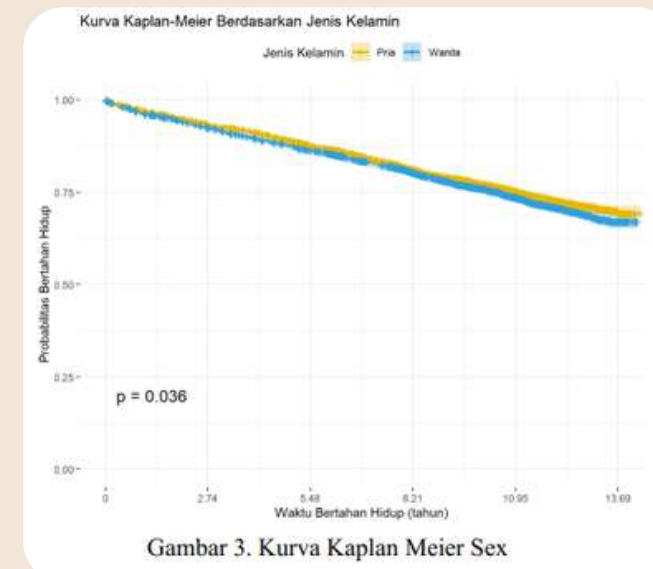
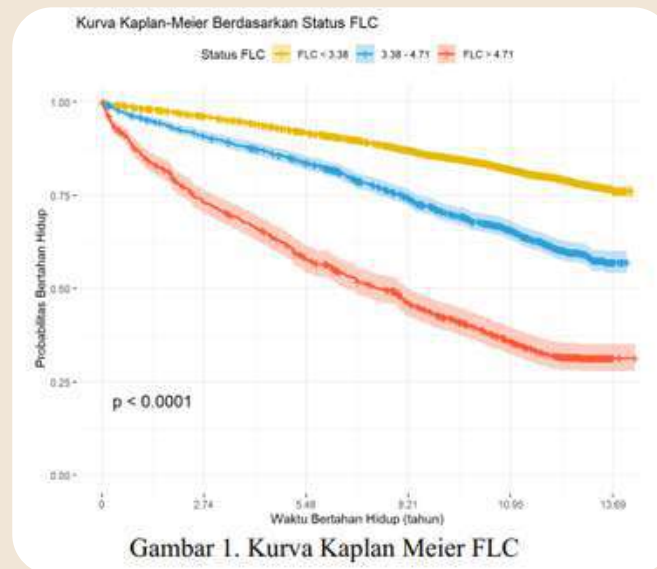
Topic :

- Full-Stack Web Development
- Database Management & ORM
- Data Pipeline & Automation
- Executive Data Visualization



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Analysis of Survival Based on Serum Free Light Chain and Other Risk Factors



This project analyzed the relationship between serum free light chain (FLC) levels and risk factors in predicting mortality using the flchain dataset from a population study in Olmsted County, Minnesota. We explored FLC as a survival biomarker alongside variables like age, gender, and MGUS status.

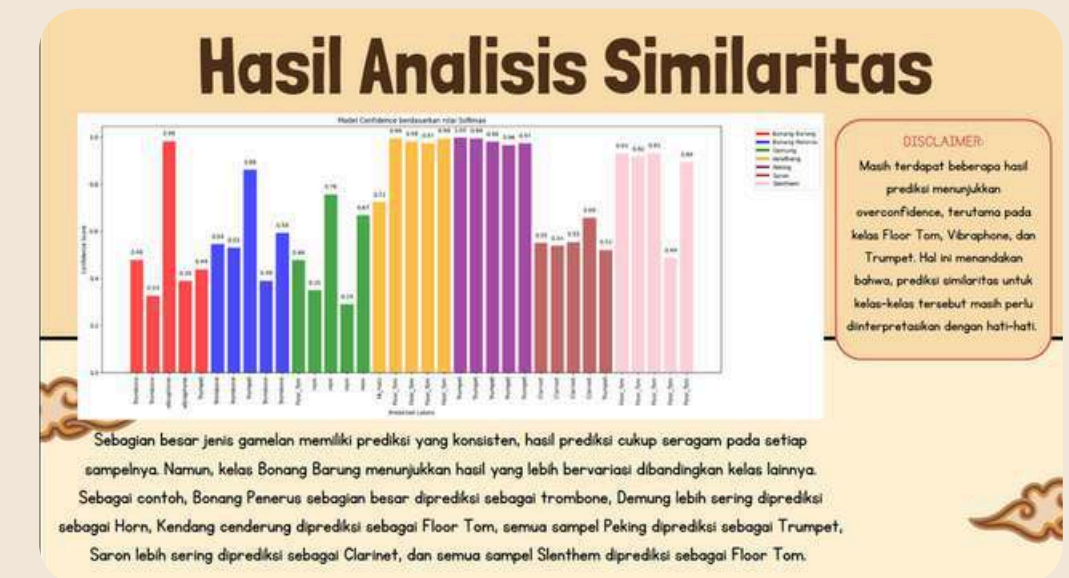
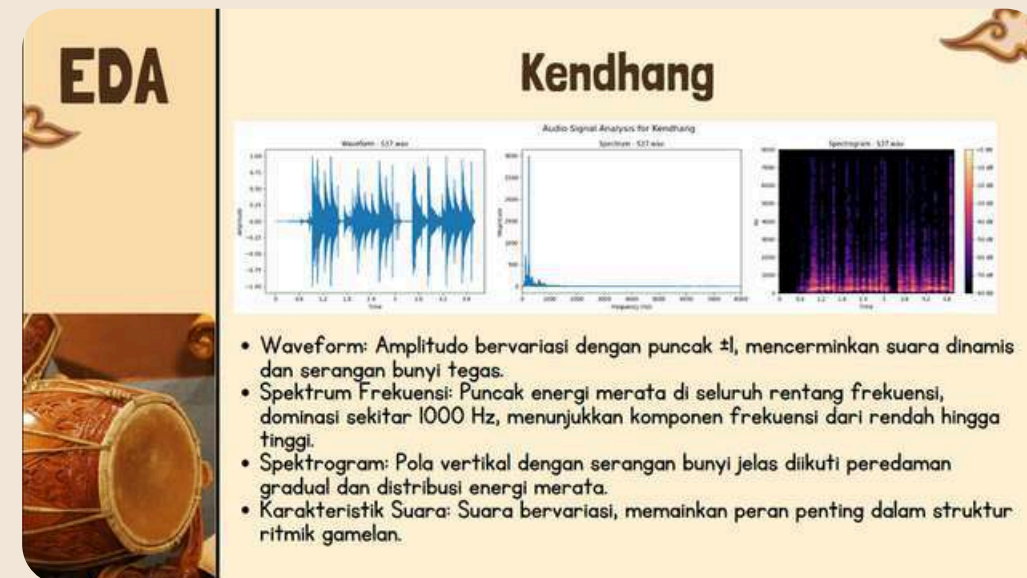
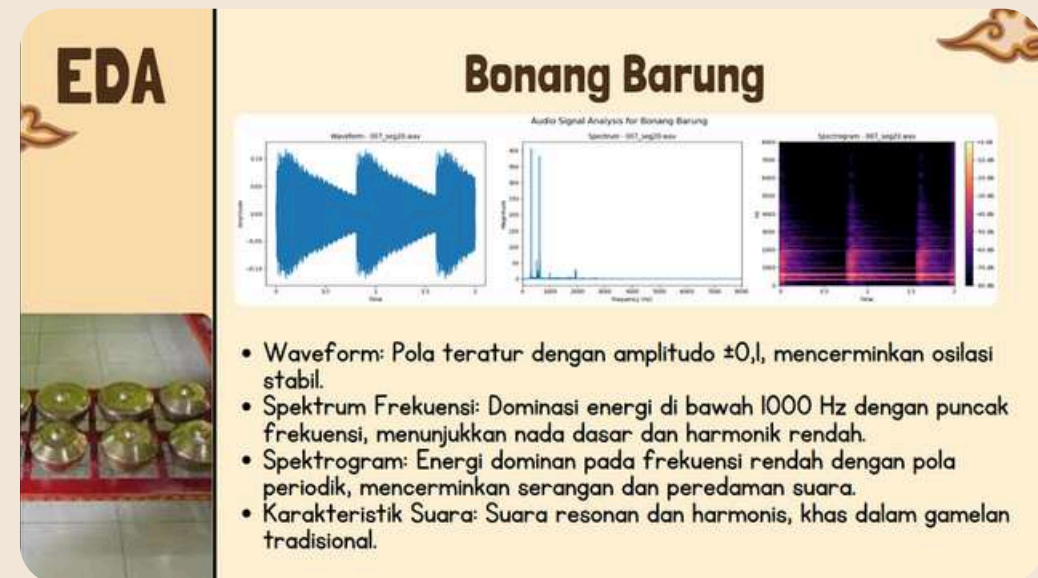
The analysis involved data cleaning, exploratory analysis, and Cox Proportional Hazard (CoxPH) modeling. To handle proportional hazards (PH) assumption violations, we applied stratified and extended Cox models, with the stratified model with interaction achieving the best fit (AIC: 30,646.04).

Topic :

- Survival Analysis
- Cox Proportional Hazard Modeling
- Stratified and Extended Cox Models
- Biomarker Analysis
- Exploratory Data Analysis (EDA)

--> [Link Projek](#) <--

Gamelan Instrument Classification Using Audio Features and Machine Learning



This project focused on classifying gamelan instruments using audio features to support the preservation of Indonesian cultural heritage. We analyzed audio recordings, extracted features using MFCC, and tested various machine learning models. Random Forest and LSTM achieved the highest accuracy at 96%, with CNN and XGBoost also performing well. I contributed to feature analysis, model development, and evaluation in this collaborative project.

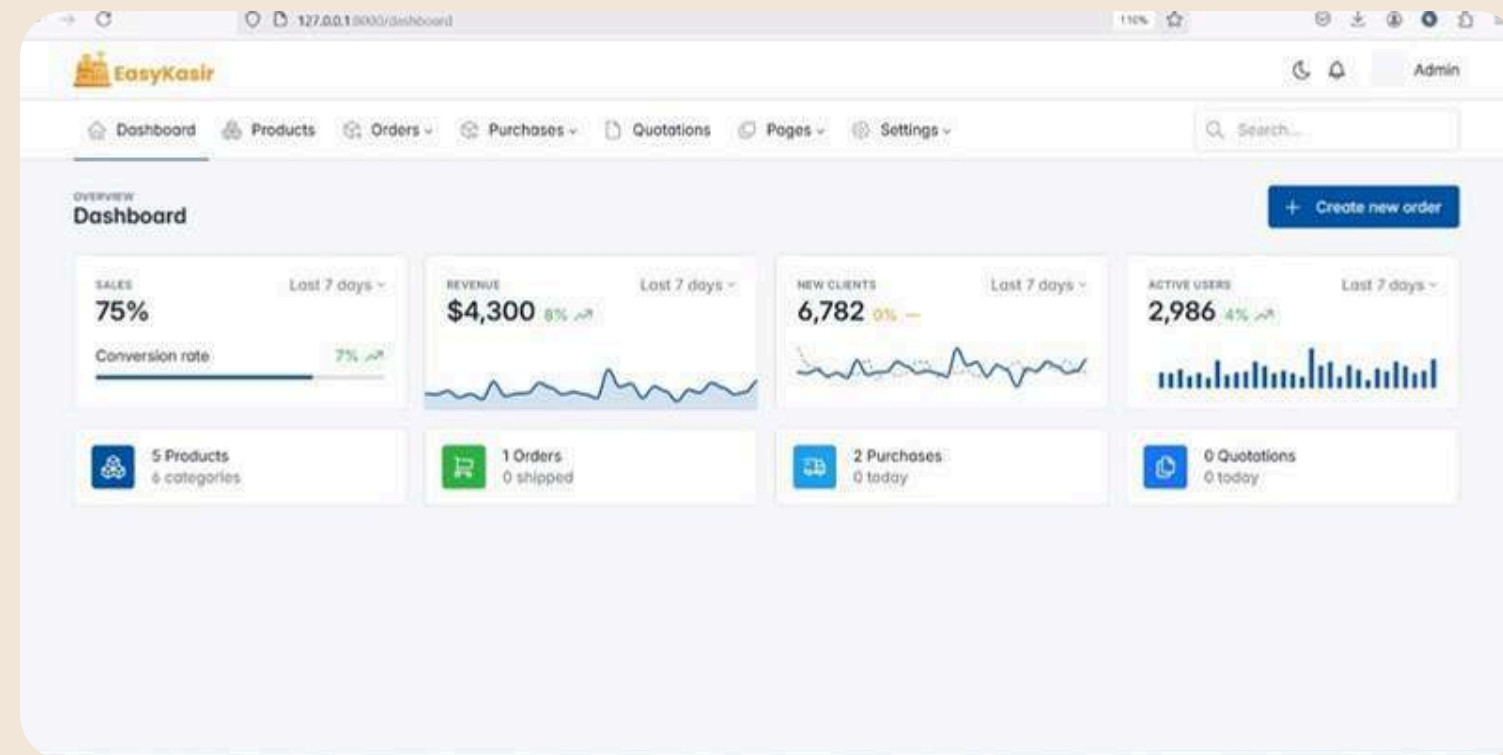
Topic :

- Audio Feature Extraction
- MFCC (Mel-Frequency Cepstral Coefficients)
- Machine Learning Classification
- Deep Learning (LSTM, CNN)
- Sound Embedding Analysis
- Exploratory Data Analysis (EDA)

--> [Link Projek](#) <--

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Implementation of Smart Cashier System Using Laravel



This project focused on developing EasyKasir, a web-based Smart Cashier System, to enhance transaction management and inventory control for small and medium-sized businesses. The system addresses the inefficiencies of traditional cashier methods, offering features like inventory tracking, sales reporting, and real-time notifications to improve operational productivity.

The system includes features for user management (admin, cashier, customer roles), inventory control, supplier management, and comprehensive sales reports. It also features a user-friendly dashboard with night mode and customizable themes for better usability.

--> [Link Projek](#) <--

Queueing System Analysis at a Barbershop – A Case Study of The Magic Barbershop, Surabaya



This project analyzes the efficiency of the queueing system at The Magic Barbershop, Keputih, Surabaya, using a Multi-Channel Single Phase Queue model (M/M/s). Observations were conducted during weekdays and weekends, both in the afternoon and evening, to evaluate variables such as arrival and service rates, waiting time, idle probability, and system utilization.

The analysis results show that evenings, especially on weekends, have the highest arrival rates and system utilization, yet remain efficient in terms of customer waiting time. The resulting recommendations include adjusting the number of staff and optimizing service time to improve customer satisfaction.

This project used Python and Google Spreadsheet for modeling and statistical analysis, providing practical insights into applying queueing theory and stochastic processes in real-world service settings.

Skills: Stochastic Methods · Statistics · Data Analysis

--> [Link Projek](#) <--

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Sentiment Analysis Review Maybank Customer in Branch



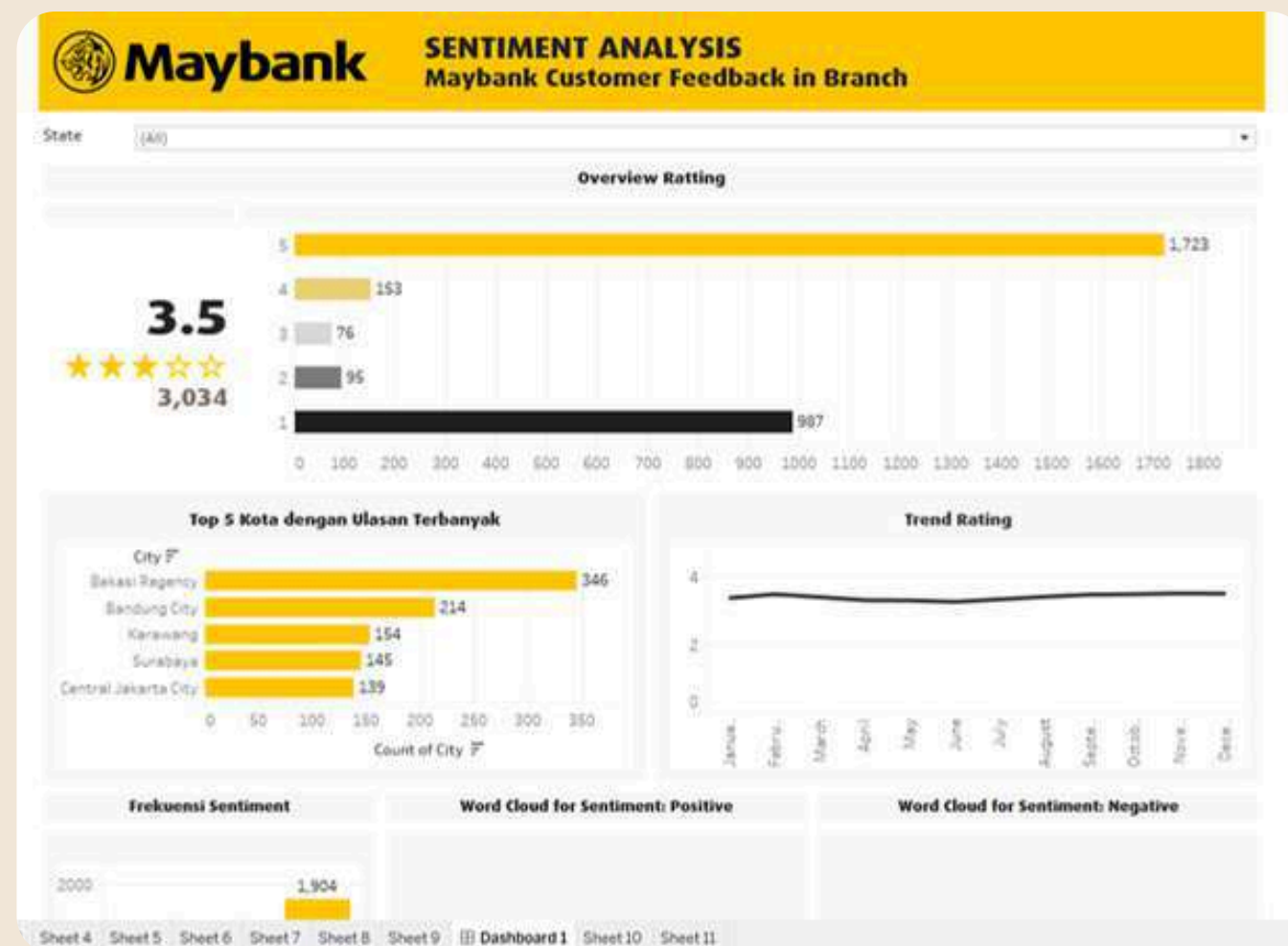
This project was part of the Data Science Consulting program (SDC22), where I joined a data analyst team to solve a real-world case study. My role in the team was as a visualization specialist, responsible for designing and building an interactive dashboard using Tableau.

The dashboard aimed to present insights from [specify data type if relevant, e.g., sales, customer satisfaction, etc.], supporting data-driven decision-making.

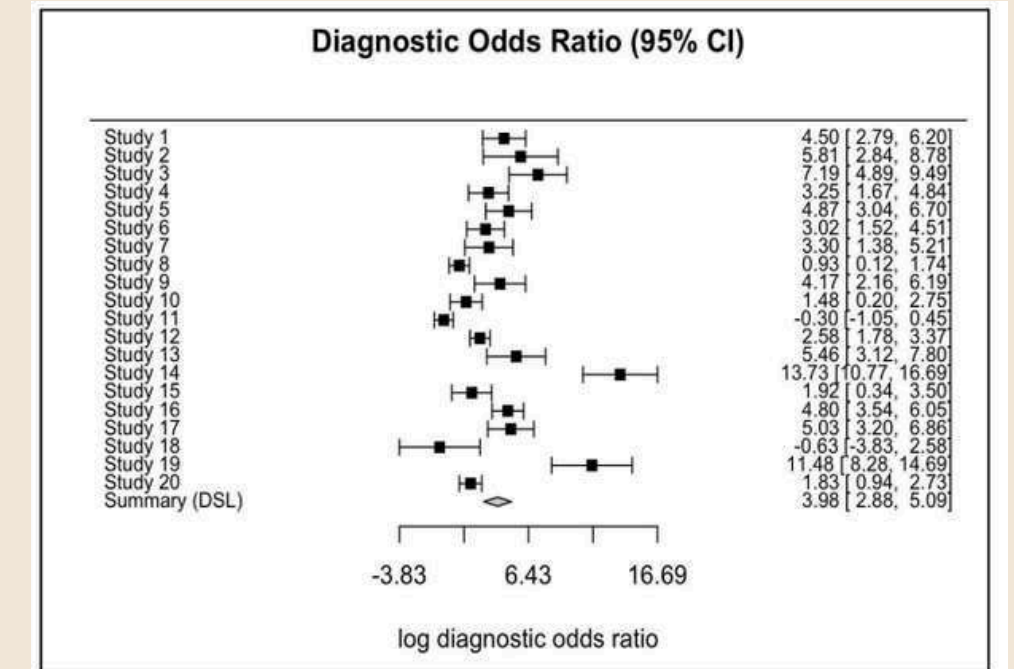
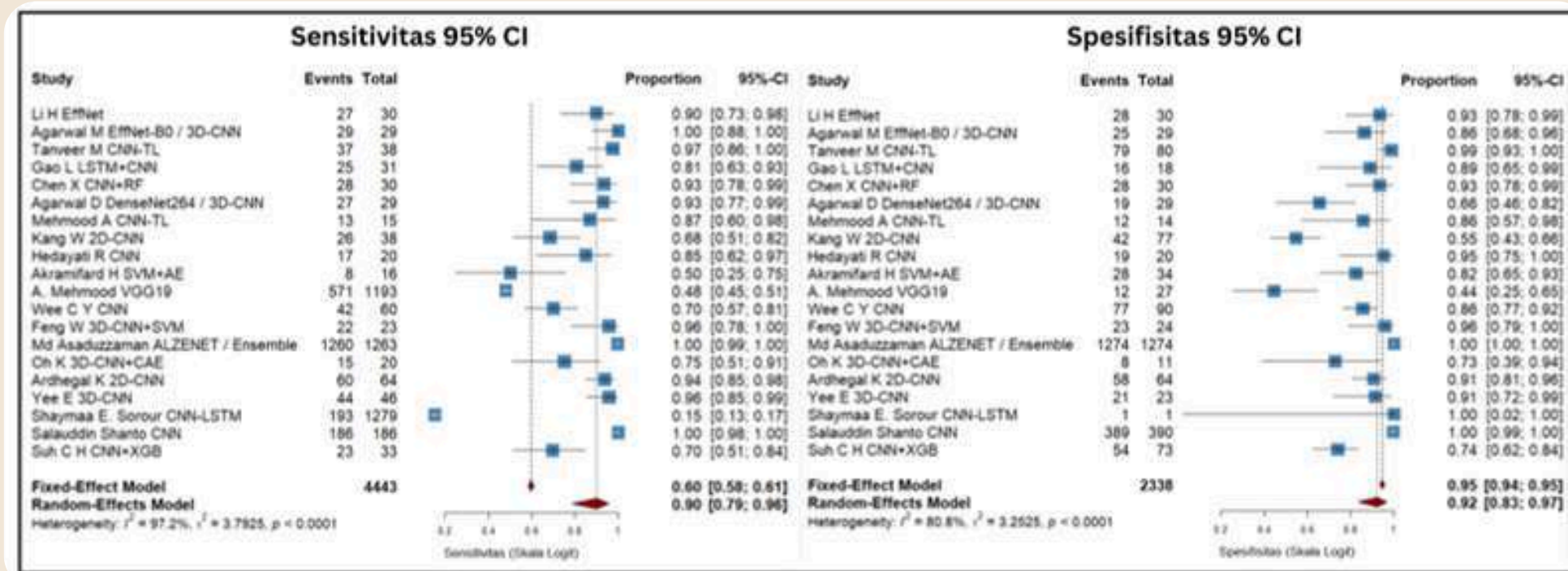
In this work, I applied effective visual design principles, built interactive layouts, and leveraged Tableau features such as dynamic filters and comparative charts to make the data easier for end users to interpret.

Skills: Data Visualization · Tableau · Data Analysis

[--> Link Projek <--](#)



Application of Machine Learning with Convolutional Neural Network Method for Alzheimer's Disease Detection: Systematic Review and Meta-Analysis



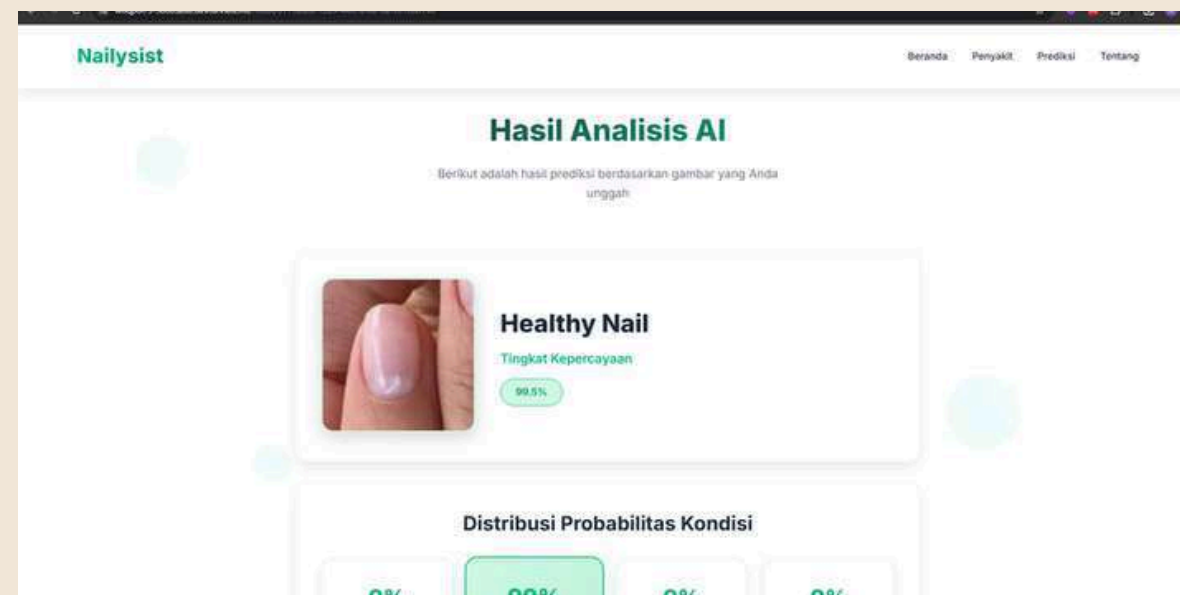
This project is a comprehensive systematic review and meta-analysis of the application of Convolutional Neural Networks (CNN) for early detection of Alzheimer's disease using brain imaging, particularly MRI.

Early detection of Alzheimer's is crucial for effective therapeutic intervention. The aim of this project is to evaluate the diagnostic performance of CNN models across various studies and to identify factors influencing their performance.

Skills: Meta-analysis · Machine Learning · Systematic Review

--> [Link Projek](#) <--

NAILYSIST: Disease Detection from Nail Images Using Machine Learning



The “NAILYSIST” project is the development of an intelligent system based on deep learning and machine learning for early detection of systemic disease indications through nail image analysis. Nail conditions can serve as important indicators of overall health, but visual diagnosis is often subjective.

This system aims to provide a fast, accurate, non-invasive, and easily accessible early diagnostic aid, especially in areas with limited medical facilities.

Skills: Machine Learning · Deep Learning · Computer Vision · Image Processing

--> [Link Projek](#) <--

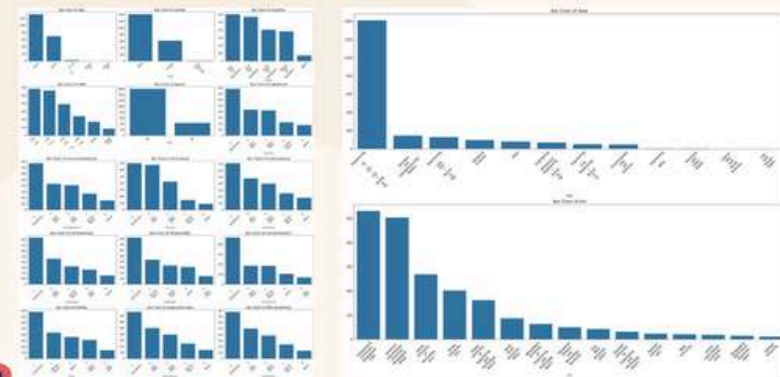
Ordinal Logistic Regression Analysis of Factors Affecting Student CGPA (Case Study in Bangladesh)

ANALISIS REGRESI LOGISTIK ORDINAL TERHADAP FAKTOR-FAKTOR YANG MEMPENGARUHI IPK MAHASISWA (STUDI KASUS DI BANGLADESH)

Kelompok 5
Analisis Data Kategorik SD-A1

CUM LAUDE
3.5

EXPLORATORY DATA ANALYSIS



UJI INDEPENDENSI

H0: Tidak ada hubungan statistik yang signifikan antara kedua variabel
H1: Ada hubungan statistik yang signifikan antara kedua variabel

Spearman (untuk variabel X ordinal)

Variabel Independen	p	p-value	Variabel Independen	p	p-value
Age	-0.048	0.038	ContCopeAcad	0.014	0.52
Gender	-0.04	0.07	ConfidentAcad	0.127	0
AcadYear	0.097	0	ThingsGoWell	0.123	0
Wolver	0.165	0	ControlIntention	0.088	0
UpsetAcad	0.01	0.647	PerfTop	0.21	0
Uncontrolab leAcad	-0.003	0.903	AngerLowGrad es	-0.018	0.419
StressAcad	0.017	0.444	DifficultiesPiling	0.023	0.3

Cramer V dan Chi-square (untuk variabel X nominal)

Variabel Independen	Chi-Square	p-value	Cramer
Uni	424.04	0	0.229
Dept	66.32	0	0.105

UJI MULTIKOLINEARITAS

Variabel	VIF	Variabel	VIF
Dept_Engineering - CS / CSE / CSE	3.78	UncontrolableAcad	2.39
Dept_Other	3.12	DifficultiesPiling	2.28
Uni_Dhaka University (DU)	2.4	ContCopeAcad	2.28
StressAcad	2.25	UpsetAcad	2.11
Dept_Engineering - EEE / ECE	2.01	Uni_Islamic University of Technology (IUT)	1.95
AngerLowGrades	1.84	Uni_Independent University (IUS)	1.74
ThingsGoWell	1.63	Uni_Other	1.56
ConfidentAcad	1.5	PerfTop	1.47
Uni_Dhaka University of Engineering (DUET)	1.45	ControlIntentions	1.39
Age	1.38	AcadYear	1.23
Gender	1.09	Wolver	1.09

PERBANDINGAN AIC DAN BIC

Kriteria	Model Awal	Model Refit
AIC	5819.74	5821.42
BIC	5943.95	5922.4

UJI KEBAIKAN MODEL

Hipotesis Uji Pearson χ^2 :

H0: Model fit

H1: Model tidak fit

Hipotesis Uji Lipsitz:

H0: Model sesuai dengan data pada semua kategori ordinal.

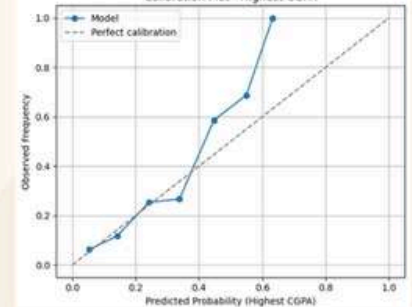
H1: Model tidak sesuai pada satu atau lebih kategori ordinal.

Nama Uji	p-value
Pearson χ^2	0.999
Lipsitz GOF χ^2	0.007

CONCORDANCE INDEX

Nilai C-Index sebesar 0,688, yang dianggap cukup baik

Calibration Plot - Highest CGPA



This project analyzes the factors influencing students' Cumulative Grade Point Average (CGPA) in Bangladesh, which serves as a key measure of academic achievement. Using an ordinal logistic regression model, the study examines demographic, academic, and psychosocial predictors of CGPA categories among 2,028 undergraduate and postgraduate students across 15 universities in Bangladesh.

Skills: Statistics · Exploratory Data Analysis · Python (Programming Language) · R (Programming Language) · Quantitative Analytics

--> [Link Projek](#) <--

Quantitative Geospatial Analysis of Regional Socio-Economic Inequality Mapping in Indonesia

ANALISIS GEOSPASIAL KUANTITATIF PEMETAAN KETIMPANGAN SOSIO-EKONOMI REGIONAL DI INDONESIA

KELOMPOK 2 - SD A2

PERBANDINGAN PENGELUARAN PER KAPITA ANTAR PROVINSI



- Terlihat jelas kesenjangan ekonomi antar provinsi.
- Pembangunan & kesejahteraan cenderung terpusat di Jawa & Barat Indonesia, kontras dengan Wilayah Timur.

PERBANDINGAN ANGKA HARAPAN HIDUP (AHH) LAKI-LAKI DAN PEREMPUAN



- **Disparitas Gender:** AHH perempuan konsisten lebih tinggi dari laki-laki di semua provinsi.
- **AHH Tertinggi:**
 - DI Yogyakarta memimpin untuk kedua gender: Laki-laki 73.7 tahun, Perempuan 77.4 tahun.
 - Menunjukkan kualitas hidup & kesehatan yang baik.
- **AHH Terendah:**
 - Didominasi wilayah Timur Indonesia (Papua Pegunungan, Papua Selatan, Papua Barat Daya).
 - Sering di bawah 65 tahun (laki-laki) dan 70 tahun (perempuan).

PEMETAAN GEOSPASIAL



This project focuses on mapping and analyzing socio-economic inequality between provinces in Indonesia for the year 2024 using a quantitative approach. Our goal is to identify and understand socio-economic disparities, uncover key factors influencing poverty levels, and reveal spatial patterns to help prioritize development planning.

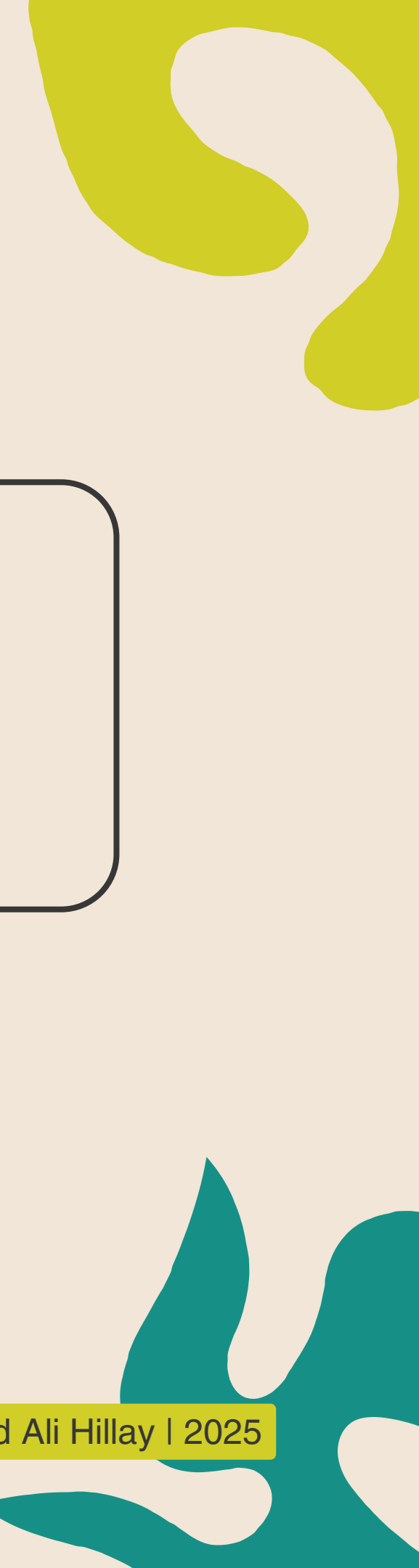
Skills: Geospatial Data · Quantitative Analytics · Exploratory Data Analysis · Data Analysis

--> [Link Proyek](#) <--

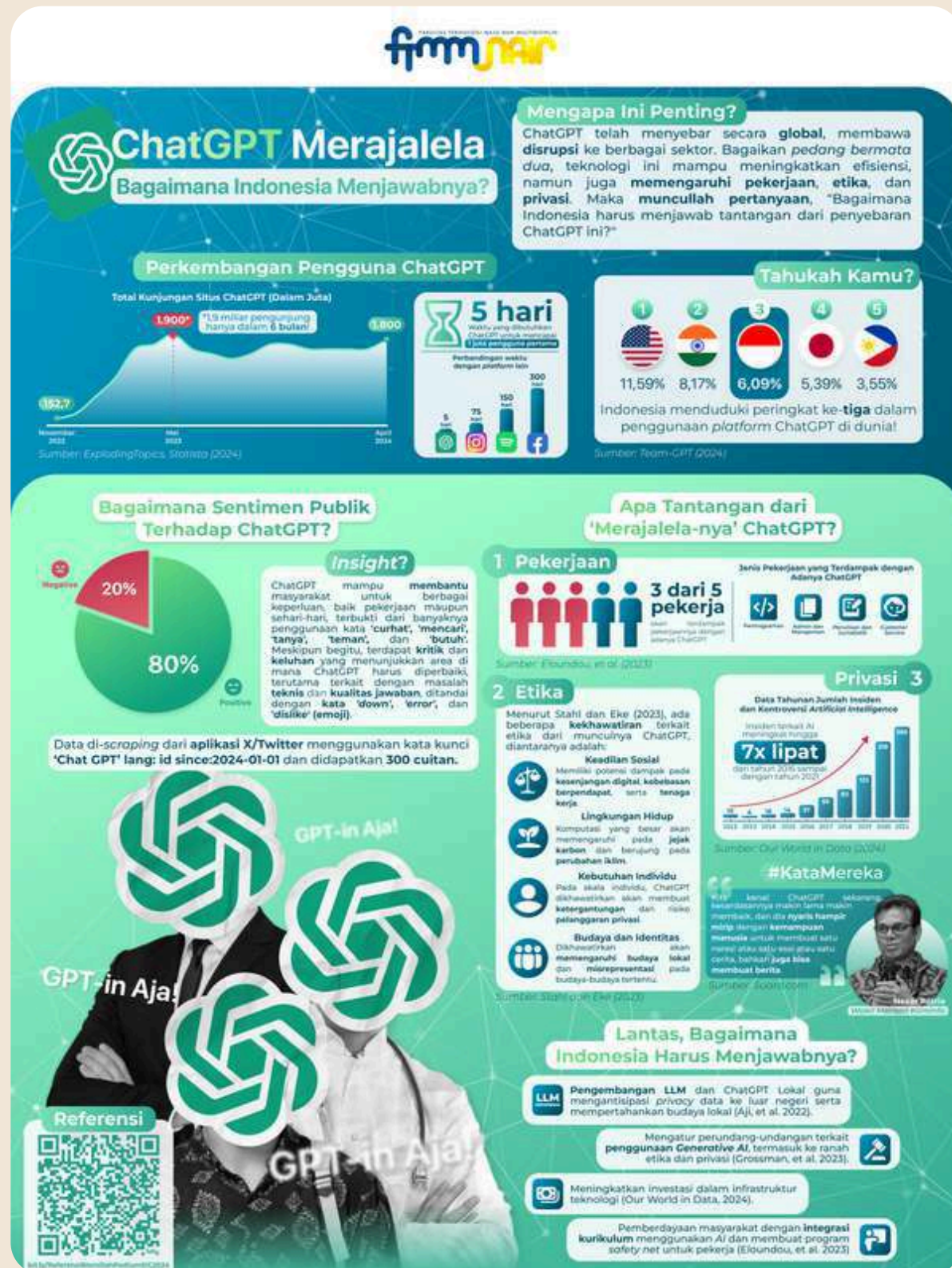


DATA VISUALIZATION PROJECT'S

2023-2026

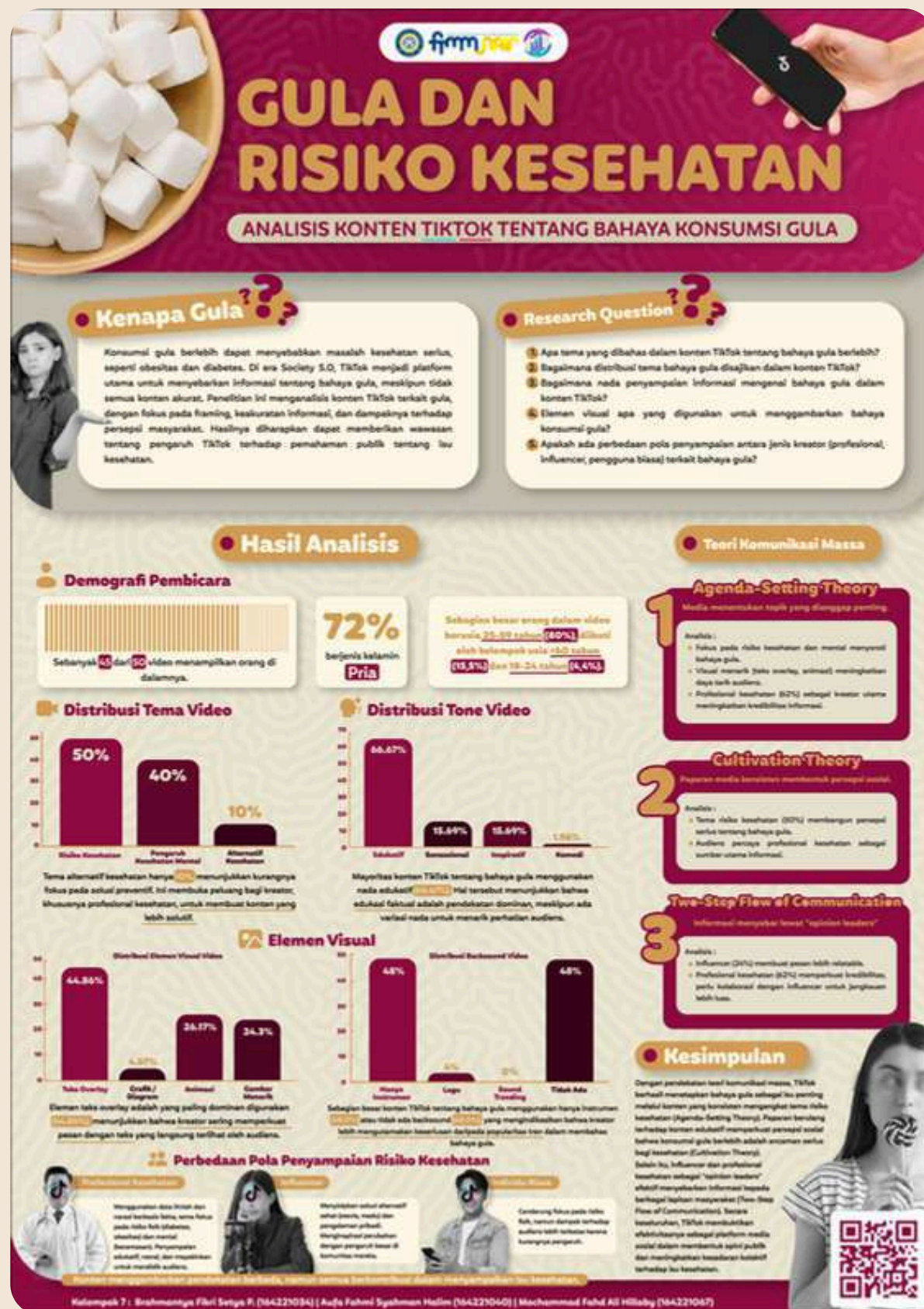


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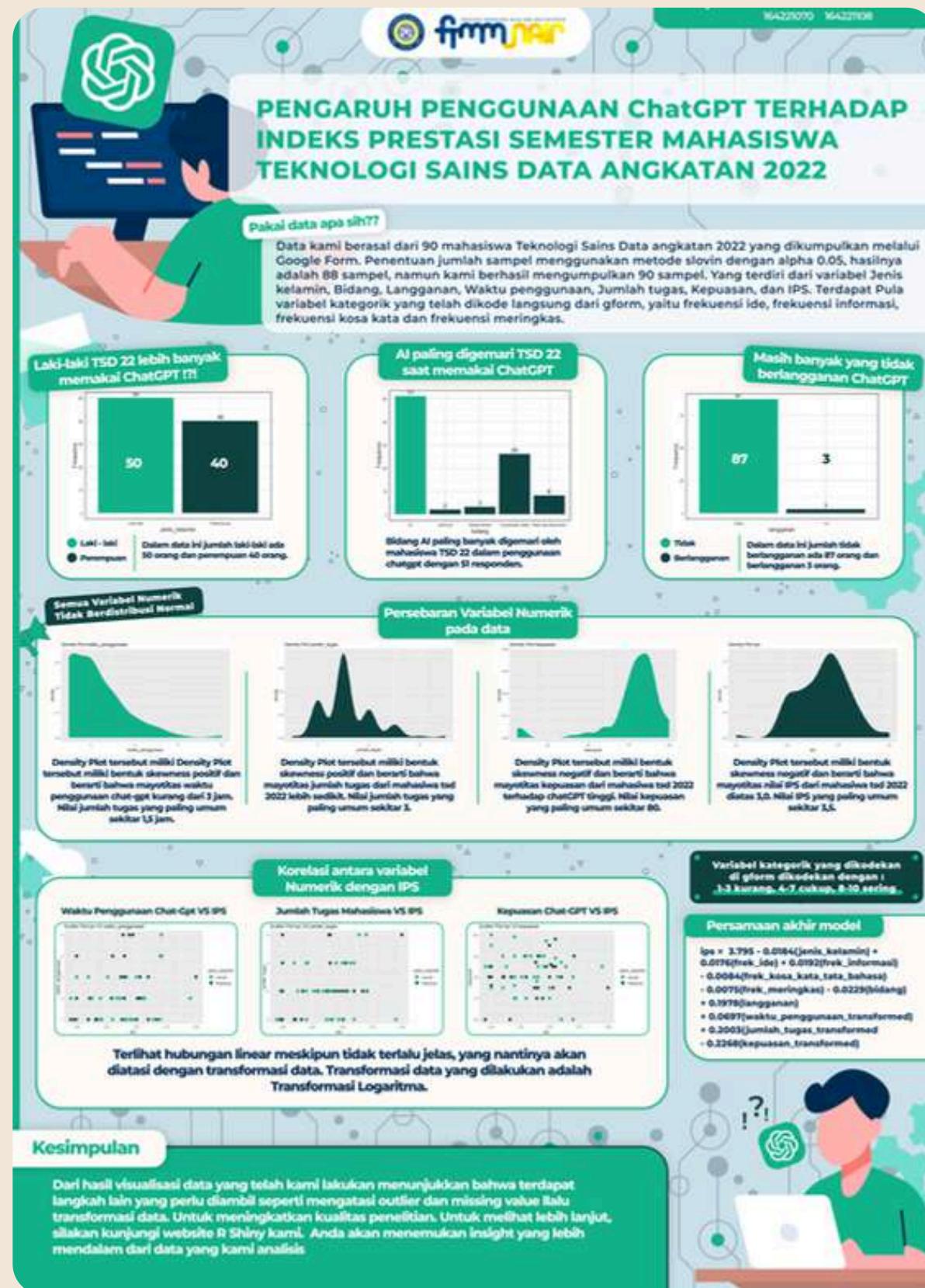
PUBLIC SENTIMENT TOWARDS CHATGPT ON SOCIAL MEDIA

This sentiment analysis visualization illustrates how the public responds to ChatGPT on social media platforms like Twitter. Words such as 'helpful' and 'search' indicate positive benefits, while 'error' and 'down' highlight areas needing improvement. I created this visualization to analyze and present the public's sentiment, showcasing both the advantages and challenges associated with ChatGPT's adoption. This visual representation helps in understanding user experiences and identifying key areas for enhancement.



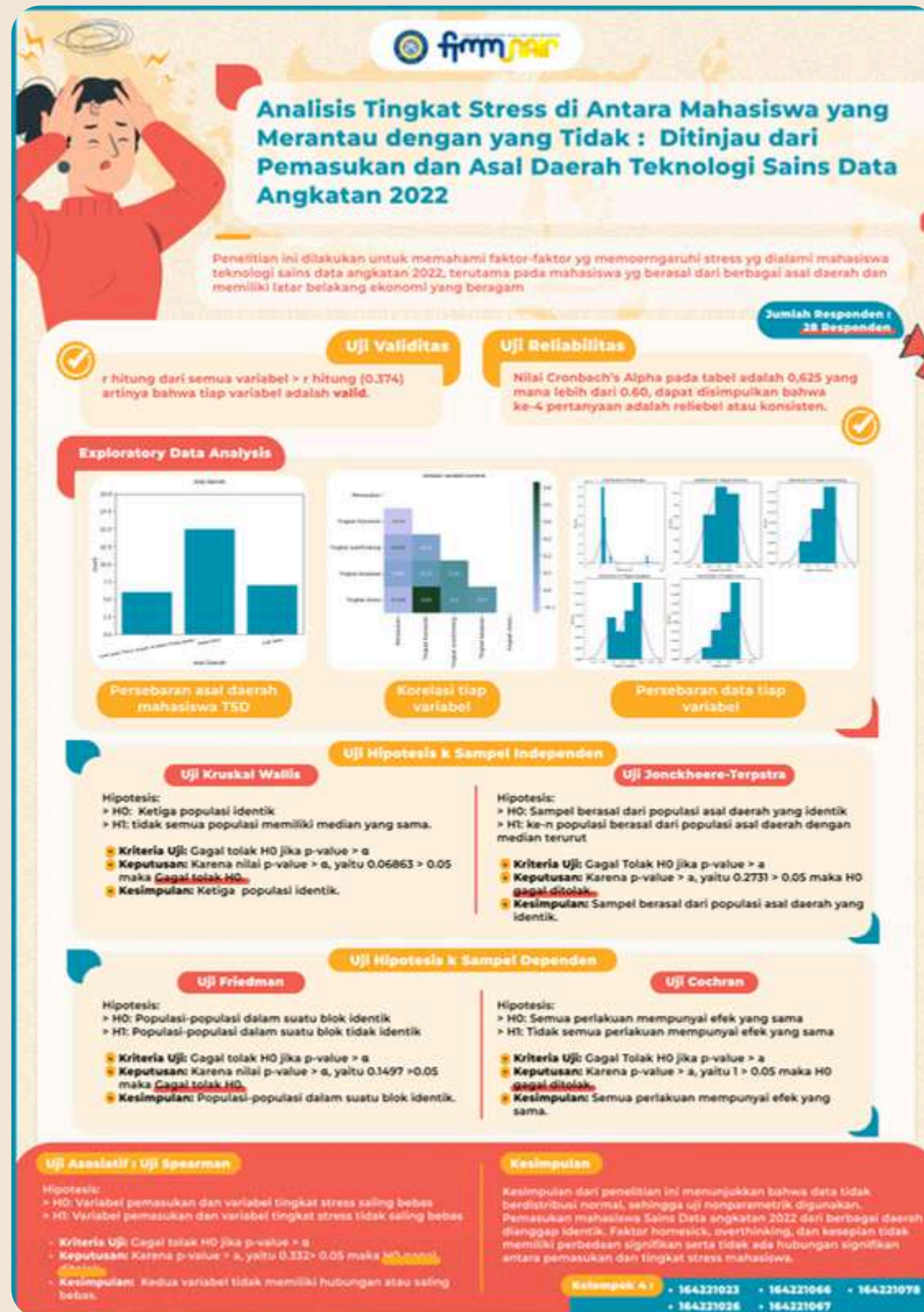
ANALYSIS OF TIKTOK CONTENT ON THE HEALTH RISKS OF SUGAR CONSUMPTION

This visualization analyzes TikTok content on the health risks of excessive sugar consumption. Key findings show a focus on women aged 25-59, with themes on health and behavior delivered in an educational tone. Videos use engaging visuals, with professionals emphasizing data and influencers promoting alternative solutions. Theories like Agenda-Setting and Cultivation highlight how TikTok shapes public perception, effectively establishing sugar risks as a critical issue and influencing societal behavior.



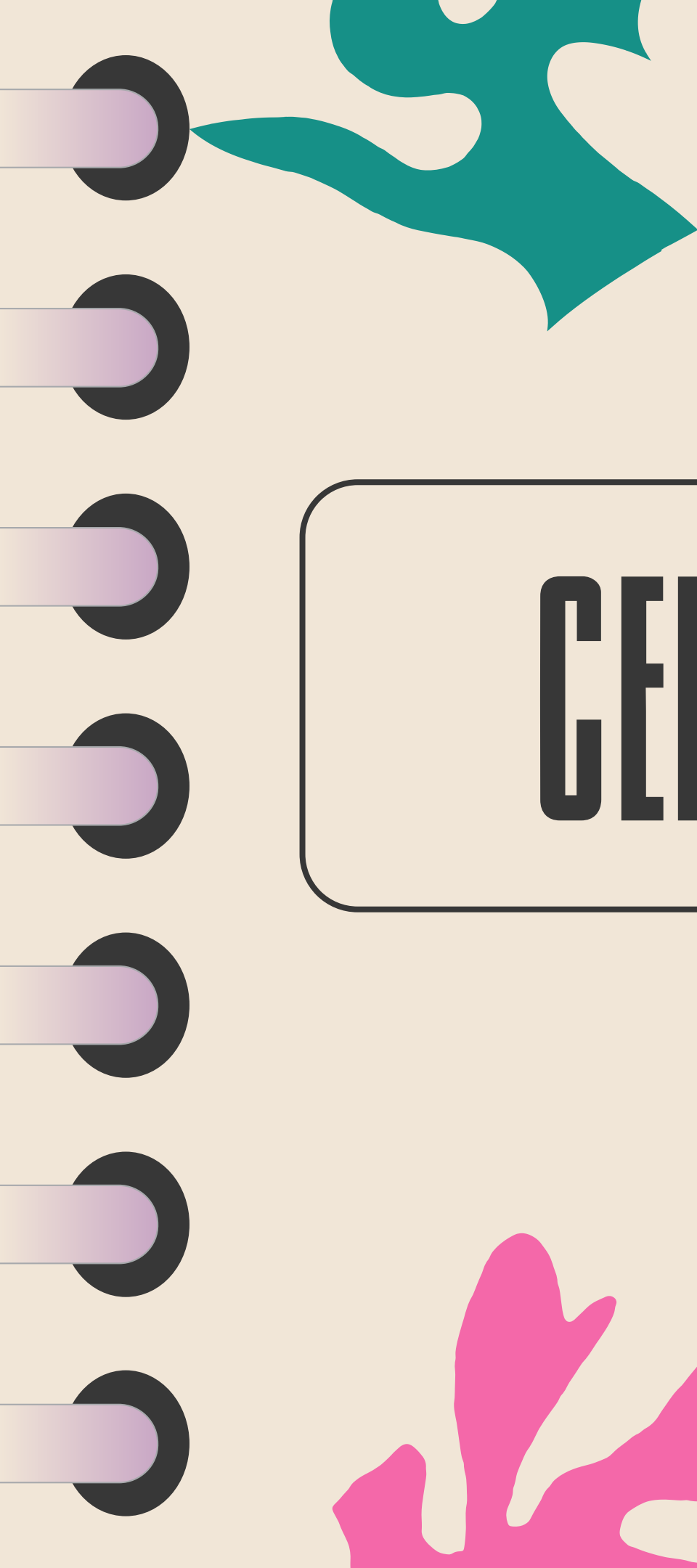
ANALYSIS OF CHATGPT USAGE IMPACT ON ACADEMIC PERFORMANCE OF DATA SCIENCE STUDENTS

This visualization analyzes ChatGPT usage among 90 Data Science students (2022 cohort) and its impact on GPA. Data includes gender, field of interest, subscription status, usage time, task frequency, satisfaction, and GPA. Findings show most students use ChatGPT for under 3 hours daily, primarily for AI tasks, with high satisfaction (avg. 80) and GPAs above 3.0. Correlation analysis links task frequency, usage time, and satisfaction to GPA, with log transformations improving accuracy. The regression model identifies key factors influencing academic performance, highlighting ChatGPT's role in student success.



ANALYSIS OF STRESS LEVELS AMONG MIGRANT AND NON-MIGRANT STUDENTS

This visualization explores stress levels among Data Science students (2022 cohort), comparing migrant and non-migrant students based on income and regional backgrounds. Data from 28 respondents was analyzed using non-parametric tests due to non-normal distribution. Key findings show no significant differences in stress levels across regions or income groups, with factors like homesickness, overthinking, and loneliness having minimal impact. Tests such as Kruskal-Wallis, Jonckheere-Terpstra, Friedman, and Spearman confirmed that stress levels and income are independent, and all treatments (stress factors) have similar effects. The study concludes that stress levels among students are consistent regardless of regional or economic differences.



CERTIFICATE & ACHIEVEMENT

2023-2026



Mochammad Fahd Ali Hillay | 2026

CERTIFICATE

119587

PUSAT BAHASA DAN MULTIBUDAYA

UNIVERSITAS AIRLANGGA

CERTIFICATE

This is to certify that

Mochammad Fahd Ali Hillaby

164221067

sat for English Language Proficiency Test

on

December 10, 2025

03251200257

Best Score Obtained :

Subjects	Score
English Listening Section	49
English Structure Section	42
English Reading Section	53
Total Score	480

Valid To : December 10, 2027



Surabaya, December 10, 2025

Head,



Lina Puryanti, S.S., M.Hum., Ph.D.
NIP. 197301311998022002

English Language Proficiency Test

CERTIFICATE



Lulusan Berprestasi Prodi
Teknologi Sains Data



Lulusan Terbaik Prodi
Teknologi Sains Data

CERTIFICATE



Data Science Course Level Basic



Data Science Course Level Intermediate

ACHIEVEMENT



Silver Medal World Invention
Competition and Exhibition 2024



Best Presentation Business and
Maritime Festival 5.0

ACHIEVEMENT



Peraih Pendanaan Program
Kreativitas Mahasiswa (PKM) 2023



Peraih Pendanaan Program
Kreativitas Mahasiswa (PKM) 2024



Peserta Pekan Ilmiah Mahasiswa
Nasional (PIMNAS) 2024

GET IN TOUCH

Feel free to reach out!



+62 812 348 55470



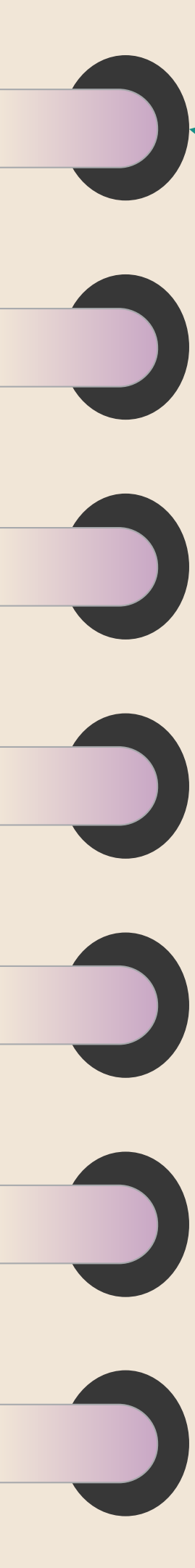
[instagram.com/fahdatauaby](https://www.instagram.com/fahdatauaby)



[linkedin.com/in/mochammad-fahd-ali-hillaby](https://www.linkedin.com/in/mochammad-fahd-ali-hillaby)



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THANK YOU